

## MODULE 6

# Cybersecurity Career Pathways

## Module Overview

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This module concludes Phase 1 by helping students understand the full landscape of cybersecurity careers and chart their personal path forward. Students will explore the wide variety of roles available in the cybersecurity industry from entry-level SOC analysts to senior architects and CISOs and understand the certifications, skills, and experience required to progress. This module also introduces students to the job market, industry expectations, and how the skills taught in this program connect directly to real employer requirements.

By completing this module, students will have a clear picture of where they are headed, what skills they need to develop, and how to begin building their professional presence in the cybersecurity industry.

## Learning Objectives

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By the end of this module, students will be able to:

1. Identify and describe at least eight distinct cybersecurity career roles across different specializations
2. Differentiate between Red Team, Blue Team, and Purple Team functions in a security organization
3. Map the skills taught in this program to specific entry-level job roles and responsibilities
4. Identify key cybersecurity certifications and explain which roles they support (CompTIA Security+, CEH, OSCP, etc.)
5. Understand the typical career progression from entry-level analyst to senior specialist or leadership roles
6. Evaluate personal strengths and interests to identify the most suitable cybersecurity specialization
7. Begin building a professional profile: resume, LinkedIn, and GitHub/portfolio strategy
8. Understand how to find and apply for cybersecurity internships and entry-level jobs

## Key Concepts & Terminology

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## The Cybersecurity Job Market

Cybersecurity is one of the fastest-growing professional fields in the world. The global shortage of cybersecurity professionals means that qualified candidates have strong job prospects across every industry from banking and healthcare to government and tech startups.

| Key Statistic                    | Detail                                                                      |
|----------------------------------|-----------------------------------------------------------------------------|
| Global talent shortage           | Over 3.5 million unfilled cybersecurity positions worldwide (2024)          |
| Average entry-level salary (USA) | \$55,000 – \$80,000 per year                                                |
| Job growth rate                  | 35% projected growth over the next 10 years (US Bureau of Labor Statistics) |
| Industries hiring                | Finance, healthcare, government, tech, retail, energy, defense              |
| Remote work availability         | High - many SOC analyst and security roles offer remote or hybrid options   |

## Red Team, Blue Team & Purple Team

The cybersecurity profession is broadly divided into offensive and defensive functions. Understanding this split is essential for identifying your career direction:

### Red Team — Offensive Security

Red teamers think and act like attackers. Their goal is to find vulnerabilities before real attackers do by simulating real-world attacks against an organization's systems, people, and processes.

Roles: Penetration Tester, Ethical Hacker, Red Team Operator, Exploit Developer

Key skills: Exploitation techniques, social engineering, scripting, OSINT, evasion

Relevant certifications: CEH, OSCP, GPEN, eCPPT

### Blue Team — Defensive Security

Blue teamers protect organizations by monitoring, detecting, and responding to threats. This is the primary focus of this training program and where most entry-level positions exist.

Roles: SOC Analyst, Incident Responder, Threat Hunter, Security Analyst

Key skills: Log analysis, SIEM tools, incident response, threat intelligence

Relevant certifications: CompTIA Security+, CompTIA CySA+, GCIA, BTL1

### Purple Team — Collaboration & Integration

Purple teamers bridge the gap between Red and Blue teams. They use offensive knowledge to improve defensive capabilities, running joint exercises and improving detection engineering.

Roles: Purple Team Analyst, Detection Engineer, Security Consultant

Key skills: Threat emulation, MITRE ATT&CK mapping, detection rule writing, automation

Relevant certifications: PNPT, GDAT, CompTIA Security+ (foundation)

## Cybersecurity Roles in Depth

### Entry-Level Roles (Target for This Program)

#### SOC Analyst — Tier 1 / Tier 2 | Typical Salary: \$55,000 – \$80,000

First line of defense in a Security Operations Center. Monitors security dashboards, investigates alerts, triages incidents, and escalates confirmed threats. Tier 2 analysts perform deeper investigation and correlation.

#### Security Analyst | Typical Salary: \$60,000 – \$85,000

Assesses the organization's security posture, performs vulnerability scans, reviews logs and reports, and recommends security improvements. Often works closely with IT teams.

#### Incident Responder (Junior) | Typical Salary: \$65,000 – \$90,000

Assists in responding to active security incidents. Helps contain threats, collects evidence, supports forensic analysis, and contributes to post-incident reports.

#### IT Security Specialist | Typical Salary: \$55,000 – \$75,000

Broad IT role with a security focus. Manages security tools, enforces policies, assists with audits, and provides first-level security support to the organization.

## Mid-Level Specializations

### **Penetration Tester / Ethical Hacker | Typical Salary: \$85,000 – \$120,000**

Conducts authorized simulated attacks against systems to identify vulnerabilities before malicious actors. Produces detailed reports with findings and remediation recommendations.

### **Threat Intelligence Analyst | Typical Salary: \$80,000 – \$110,000**

Researches threat actor groups, attack campaigns, and emerging vulnerabilities. Produces intelligence reports and feeds indicators of compromise (IOCs) into security tools.

### **Threat Hunter | Typical Salary: \$90,000 – \$120,000**

Proactively searches for hidden threats that automated tools have missed. Uses hypothesis-driven investigation techniques and advanced log analysis to uncover stealthy attackers.

### **Digital Forensics Analyst | Typical Salary: \$75,000 – \$105,000**

Investigates cybercrime and security incidents by examining digital evidence. Recovers deleted data, analyzes malware, reconstructs attack timelines, and supports legal proceedings.

### **Security Engineer | Typical Salary: \$100,000 – \$140,000**

Designs, builds, and maintains security infrastructure including firewalls, SIEM systems, EDR platforms, and identity management solutions.

### **Cloud Security Engineer | Typical Salary: \$110,000 – \$150,000**

Secures cloud infrastructure on platforms such as AWS, Azure, and Google Cloud. Manages identity and access, monitors cloud environments, and enforces cloud security policies.

## Advanced & Leadership Roles

### **Security Architect | Typical Salary: \$130,000 – \$170,000**

Designs enterprise-wide security frameworks, defines standards, and ensures that all systems are built with security in mind. Works at the intersection of technical design and business strategy.

### **SOC Manager | Typical Salary: \$110,000 – \$150,000**

Leads a Security Operations Center team. Manages analysts, incident response processes, SLAs, reporting, and continuous improvement of detection capabilities.

## Chief Information Security Officer (CISO) | Typical Salary: \$180,000 – \$300,000+

Executive leader responsible for the entire organizational security strategy, budget, team, and risk posture. Reports directly to the CEO or Board. Requires deep technical and business experience.

## Cybersecurity Certifications Roadmap

Certifications validate skills and significantly improve employability. The table below maps certifications to career stages and roles:

| Certification                  | Level              | Best For                                                    | Estimated Cost        |
|--------------------------------|--------------------|-------------------------------------------------------------|-----------------------|
| CompTIA Security+              | Entry              | All blue team roles, SOC Analyst, Security Analyst          | \$392 (exam only)     |
| CompTIA Network+               | Entry              | SOC Analyst, Network Security roles (take before Security+) | \$338 (exam only)     |
| CompTIA CySA+                  | Intermediate       | SOC Analyst Tier 2, Threat Hunter, Incident Responder       | \$392 (exam only)     |
| Blue Team Labs Online (BTL1)   | Entry-Intermediate | SOC Analyst, Incident Responder - hands-on lab focused      | \$200 (course + exam) |
| Certified Ethical Hacker (CEH) | Intermediate       | Penetration Tester, Red Team entry-level                    | \$950–\$1,199         |
| eJPT (eLearnSecurity)          | Entry (Offensive)  | Penetration Tester beginner, practical skills               | \$200                 |
| OSCP (Offensive Security)      | Advanced           | Senior Penetration Tester, Red Team Operator                | \$1,499               |
| GIAC GCIH                      | Intermediate       | Incident Responder, SOC Tier 2/3                            | \$979 (exam only)     |
| AWS/Azure Security Specialty   | Advanced           | Cloud Security Engineer                                     | \$300 (exam only)     |
| CISSP                          | Advanced           | Security Architect, CISO, Senior Manager                    | \$749 (exam only)     |

## Career Progression Pathway

Understanding the typical career progression helps students set realistic goals and plan their learning journey beyond this program:

- 1 Entry Level (0–2 years)** – SOC Analyst Tier 1, IT Security Specialist, Security Analyst. Focus: building fundamentals, earning Security+, gaining hands-on tool experience.
- 2 Junior Specialist (2–3 years)** – SOC Analyst Tier 2, Junior Incident Responder, Junior Penetration Tester. Focus: deepening specialization, earning intermediate certifications (CySA+, BTL1, eJPT).
- 3 Mid-Level (3–6 years)** – Incident Responder, Threat Hunter, Penetration Tester, Forensics Analyst. Focus: leading investigations, mentoring juniors, earning advanced certifications (GCIH, OSCP, CEH).
- 4 Senior Specialist (6–10 years)** – Senior Security Engineer, Senior Threat Intelligence Analyst, Senior Penetration Tester. Focus: system design, team leadership, architecture decisions.
- 5 Leadership (10+ years)** – Security Architect, SOC Manager, CISO, Director of Security. Focus: strategy, budgets, executive communication, organizational risk management.